ÁLVARO FRANCISCO GIL

Software Engineer | Generative AI | Research

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STATEMENT

Alvaro Francisco Gil is a Software Engineer with a strong passion for learning and a huge ambition. His lifelong goal is to be one of the leading Computer Science experts globally, to advance the boundaries of comprehensive knowledge and build a sustainable world powered by technology.

Exploiting his innate curiosity, he has devoured books of all disciplines: literature, psychology, mathematics, philosophy, entrepreneurship... Always driven by the desire to understand the mystery of the human being, his brain, and his role in society. Álvaro believes the best way to fully comprehend something is to build it from scratch, and this is exactly the origin of his obsession with Artificial Intelligence. The most remarkable aspect of his education has been his interest in learning from different perspectives, traveling to new cultures. He started studying at UPM, did an Erasmus at PK, and conducted research at MIT.

Apart from his fascination with theory, he understands that every advance must be properly brought to society. He conceives this process as an art, the art of business. He enjoys psychoanalyzing the client or even his co-workers, in order to establish effective communication, creating a climate of mutual benefit that leads to happiness. Although in his history there have been some entrepreneurial ventures, he now prefers to join companies already formed, with a perspective of exponential growth and a high impact on society.

EXPERIENCE

Chatbot Engineer
NorthStar Earth & Space
04/2025 - Present | Luxembourg, Luxembourg

Integration of a chatbot into a SaaS platform for satellite monitoring analytics.

- Google ADK
- Model Context Protocol
- Node.js
- Kubernetes
- AWS
- CI/CD

Generative AI Engineer
Universidad Politécnica de Madrid
03/2024 - 04/2025 | Madrid, Spain

Tuning of Variational Autoencoders (VAEs) to generate physically realistic trajectories.

- Generative AI
- Docker
- Astrodynamics

Deep Learning Researcher
Massachusetts Institute of Technology (MiT)
08/2023 - 03/2024 | Cambridge, MA

AI-driven Electrical Signal Analysis for Bio-sensor Development.

- PyTorch
- Signal Processing
- Scientific Writing

Machine Learning Engineer
Habber Tec España
03/2023 - 08/2023 | Madrid, Spain

Consultancy project to forecast failures in electricity transformers.

- Machine Learning
- Time Series
- Documentation

Computational Analyst
Universidad Politécnica de Madrid
06/2022 - 06/2023 | Madrid, Spain

Study on social dynamics to enhance collaboration through data analysis.

- AI for Emotion Recognition
- Social Network Analysis
- Diarization
- Speach to Text

EDUCATION

Master in Language Technologies
Universidad Nacional de Educación a Distancia
09/2024 - 06/2025

Engineer's degree: Software Engineering
Universidad Politécnica de Madrid
09/2018 - 06/2023

Computer Science, Erasmus
Politechnika Krakowska
09/2020 - 03/2021

PUBLICATIONS

Generative Design of Periodic Orbits in the Restricted Three-Body Problem
SPAICE 2024 (organized by ESA)
A.F.Gil,
10/2024 | https://zenodo.org/records/13885649

First paper that ever uses Generative AI in Astrodynamics for orbit generation.

Can Plants Perceive Human Gestures? Using AI to Track Eurythmic Human–Plant Interaction
MDPI Sensors (Q1)
A.F.Gil,
02/2024 | https://www.mdpi.com/2313-7673/9/5/290

Exploration on how AI can reveal plant responses by analyzing electrical signals.

Comparing ethical values and business performance through SNA and emotion analysis
Sunbelt 2023
A.Fronzetti, P.Gloor, A.F.Gil, J.Garbajosa
06/2023 | https://www.insna.org/

Analysis of the importance of collaboration and its impact on overall performance.

LANGUAGES

Spanish
Native

English
Proffesional

Python
Proffesional

SKILLS

Google ADK | Model Context Protocol | PyTorch | Docker | Kubernetes | AWS | Firebase | Node.js | Langchain | REST APIs | Prompting | Transformers | NLP | Machine Learning | Git

AWARDS

- **MIT Research Fellowship 2024**
- **Eelisa Communities Grant 2022**